

Final Exam

PhD Macro (Spring 2026)

Instructions. Show steps. Partial credit for correct logic and intermediate steps.

Question 1: Human capital accumulated using resources (goods) [35 pts]

Consider a planner economy with two accumulable stocks: physical capital K_t and human capital H_t . Output is produced with constant returns to the *vector* (K, H) :

$$Y_t = AK_t^\alpha H_t^{1-\alpha}, \quad \alpha \in (0, 1), \quad A > 0.$$

The resource constraint is

$$C_t + I_t^K + I_t^H = Y_t.$$

Stocks evolve as

$$K_{t+1} = (1 - \delta)K_t + I_t^K, \quad H_{t+1} = (1 - \delta)H_t + I_t^H,$$

with the *same* depreciation rate $\delta \in (0, 1)$. Preferences are CRRA:

$$\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma} - 1}{1 - \sigma}, \quad \beta \in (0, 1), \quad \sigma > 0.$$

- (a) **Static allocation across K vs H (one-period condition).** [12 pts]

Show that along an interior optimum (both $I_t^K > 0$ and $I_t^H > 0$), the planner must satisfy

$$MPK_t = MPH_t,$$

where $MPK_t \equiv \partial Y_t / \partial K_t$ and $MPH_t \equiv \partial Y_t / \partial H_t$. Use this to show the **balanced-growth ratio** is constant:

$$\frac{K_t}{H_t} = \frac{\alpha}{1 - \alpha}.$$

- (b) **Balanced growth rate.** [13 pts]

Let $\gamma \equiv C_{t+1}/C_t = K_{t+1}/K_t = H_{t+1}/H_t$ be the common gross growth rate on a balanced growth path (BGP). Show that

$$\gamma^\sigma = \beta(MPK + 1 - \delta),$$

where MPK is the (constant) marginal product of capital on the BGP.

- (c) **Numerical.** [10 pts]

Take $\beta = 0.96$, $\sigma = 2$, $\delta = 0.08$, $A = 0.25$, $\alpha = 0.40$. Compute: (i) the BGP ratio K/H ; (ii) MPK on the BGP; and (iii) the implied γ .

Solution. (a) With one unit of resources, the planner can increase next period K_{t+1} by 1 (via I_t^K) or increase H_{t+1} by 1 (via I_t^H). At an interior optimum, the marginal value of investing in either stock must be equal. With identical depreciation and same investment good, this implies the gross marginal payoff is equal:

$$MPK_t + 1 - \delta = MPH_t + 1 - \delta \Rightarrow MPK_t = MPH_t.$$

Now compute marginal products:

$$MPK_t = \alpha AK_t^{\alpha-1} H_t^{1-\alpha}, \quad MPH_t = (1 - \alpha) AK_t^\alpha H_t^{-\alpha}.$$

Set $MPK_t = MPH_t$:

$$\alpha AK^{\alpha-1} H^{1-\alpha} = (1 - \alpha) AK^\alpha H^{-\alpha} \Rightarrow \alpha H = (1 - \alpha) K \Rightarrow \frac{K}{H} = \frac{\alpha}{1 - \alpha}.$$

(b) Euler equation for investing in K (standard planner Euler) is:

$$C_t^{-\sigma} = \beta C_{t+1}^{-\sigma} (MPK_{t+1} + 1 - \delta).$$

On a BGP, $C_{t+1}/C_t = \gamma$ and $MPK_{t+1} = MPK$ is constant because K/H is constant. Thus

$$\left(\frac{C_{t+1}}{C_t}\right)^\sigma = \beta(MPK + 1 - \delta) \Rightarrow \gamma^\sigma = \beta(MPK + 1 - \delta).$$

(c) (i) $K/H = \alpha/(1 - \alpha) = 0.40/0.60 = 2/3$.

(ii) Write MPK in terms of $x \equiv K/H$:

$$MPK = \alpha AK^{\alpha-1} H^{1-\alpha} = \alpha A(K/H)^{\alpha-1} = \alpha A x^{\alpha-1}.$$

Here $x = 2/3$ and $\alpha - 1 = -0.60$, so

$$x^{\alpha-1} = (2/3)^{-0.60} = (3/2)^{0.60} \approx e^{0.60 \ln(1.5)} \approx e^{0.60 \cdot 0.4055} \approx e^{0.2433} \approx 1.275.$$

Hence

$$MPK \approx 0.40 \cdot 0.25 \cdot 1.275 = 0.1275.$$

(iii) $\gamma^\sigma = \beta(MPK + 1 - \delta) = 0.96(0.1275 + 0.92) = 0.96(1.0475) = 1.0056$. With $\sigma = 2$, $\gamma = \sqrt{1.0056} \approx 1.0028$, i.e. about 0.28% growth per period.

Question 2: Lucas human capital (time allocation) [30 pts]

Consider a Lucas-style human capital model with no physical capital. Human capital evolves according to

$$H_{t+1} = (1 - \delta_H)H_t + \phi u_t H_t, \quad \delta_H \in (0, 1), \phi > 0,$$

where $u_t \in [0, 1]$ is the fraction of time spent accumulating human capital (education/training). Goods production uses effective labor $(1 - u_t)H_t$:

$$Y_t = A(1 - u_t)H_t, \quad A > 0.$$

Resource constraint: $C_t = Y_t$. Preferences are log utility:

$$\sum_{t=0}^{\infty} \beta^t \log C_t, \quad \beta \in (0, 1).$$

(a) **Constant- u representation.**

[12 pts]

Suppose $u_t = u$ is constant over time. Show that

$$C_t = A(1 - u)H_t, \quad \frac{H_{t+1}}{H_t} = 1 - \delta_H + \phi u \equiv \gamma_H(u).$$

Conclude that consumption grows at gross rate $\gamma_C(u) = \gamma_H(u)$.

(b) **Choose the optimal u (calculation).**

[18 pts]

Under constant u , show the lifetime utility can be written as

$$\sum_{t \geq 0} \beta^t \log C_t = \text{constant} + \frac{1}{1 - \beta} \log(1 - u) + \frac{\beta}{(1 - \beta)^2} \log(\gamma_H(u)),$$

up to terms that do not depend on u . Derive the FOC characterizing the optimal u^* :

$$\frac{1}{1 - u^*} = \frac{\beta}{1 - \beta} \cdot \frac{\phi}{1 - \delta_H + \phi u^*}.$$

Then compute u^* when $\beta = 0.95$, $\delta_H = 0.02$, $\phi = 0.10$.

Solution. (a) With constant u , output and consumption are $C_t = A(1 - u)H_t$. Human capital law of motion becomes $H_{t+1} = (1 - \delta_H + \phi u)H_t$, so $\gamma_H(u) = H_{t+1}/H_t = 1 - \delta_H + \phi u$. Since $C_t \propto H_t$ with constant proportionality $A(1 - u)$, consumption grows at the same rate: $\gamma_C(u) = \gamma_H(u)$.

(b) Write $\log C_t = \log A + \log(1 - u) + \log H_t$. Also $\log H_t = \log H_0 + t \log \gamma_H(u)$. Therefore

$$\sum_{t=0}^{\infty} \beta^t \log C_t = \sum_{t=0}^{\infty} \beta^t [\log A + \log(1 - u) + \log H_0 + t \log \gamma_H(u)].$$

Use $\sum_{t \geq 0} \beta^t = 1/(1 - \beta)$ and $\sum_{t \geq 0} t\beta^t = \beta/(1 - \beta)^2$. Terms $\log A, \log H_0$ do not depend on u , so (up to constants):

$$\frac{1}{1 - \beta} \log(1 - u) + \frac{\beta}{(1 - \beta)^2} \log \gamma_H(u).$$

Differentiate w.r.t. u :

$$-\frac{1}{1 - \beta} \cdot \frac{1}{1 - u} + \frac{\beta}{(1 - \beta)^2} \cdot \frac{1}{\gamma_H(u)} \cdot \phi = 0.$$

Multiply by $(1 - \beta)^2$:

$$-(1 - \beta) \frac{1}{1 - u} + \beta \frac{\phi}{\gamma_H(u)} = 0 \Rightarrow \frac{1}{1 - u} = \frac{\beta}{1 - \beta} \cdot \frac{\phi}{1 - \delta_H + \phi u}.$$

Now plug numbers: $\beta/(1 - \beta) = 0.95/0.05 = 19$. So

$$\frac{1}{1 - u} = 19 \cdot \frac{0.10}{0.98 + 0.10u} = \frac{1.9}{0.98 + 0.10u}.$$

Invert:

$$1 - u = \frac{0.98 + 0.10u}{1.9}.$$

Multiply by 1.9:

$$1.9 - 1.9u = 0.98 + 0.10u \Rightarrow 0.92 = 2.0u \Rightarrow u^* = 0.46.$$

Question 3: Production heterogeneity — aggregation proof and distribution effects [35 pts]

There is a continuum of firms indexed by idiosyncratic productivity $\varepsilon > 0$, distributed with cdf μ . Each firm has technology

$$y(\varepsilon) = z \varepsilon k(\varepsilon)^\alpha n(\varepsilon)^\nu, \quad \alpha, \nu \in (0, 1), \quad \alpha + \nu < 1,$$

and rents capital and hires labor at common prices $r + \delta$ and w . Firm profits:

$$\pi(\varepsilon) = z \varepsilon k^\alpha n^\nu - (r + \delta)k - wn.$$

Aggregate inputs are $K = \int k(\varepsilon) d\mu(\varepsilon)$ and $N = \int n(\varepsilon) d\mu(\varepsilon)$.

- (a) **Firm demands scale with productivity.** [12 pts]
Show that optimal inputs can be written as

$$k(\varepsilon) = \kappa_k \varepsilon^{\frac{1}{1-\alpha-\nu}}, \quad n(\varepsilon) = \kappa_n \varepsilon^{\frac{1}{1-\alpha-\nu}},$$

for constants $\kappa_k, \kappa_n > 0$ that depend only on $(z, r + \delta, w, \alpha, \nu)$, not on ε .

- (b) **Aggregation and the productivity index $A(\mu)$.** [13 pts]
Define the moment

$$M(\mu) \equiv \int \varepsilon^{\frac{1}{1-\alpha-\nu}} d\mu(\varepsilon).$$

Show that aggregate output can be written as

$$Y = z K^\alpha N^\nu A(\mu), \quad A(\mu) \equiv \left(M(\mu) \right)^{1-\alpha-\nu}.$$

(Interpret $A(\mu)$ as an “endogenous TFP” term.)

(c) **How distribution changes affect aggregate productivity.** [10 pts]

Let $\alpha = 0.30$, $\nu = 0.50$, so $1 - \alpha - \nu = 0.20$. Consider two distributions with the same mean $\mathbb{E}[\varepsilon] = 1$:

$$\mu_1 : \varepsilon = 1 \text{ w.p. } 1, \quad \mu_2 : \varepsilon = 0.5 \text{ w.p. } 0.5, \varepsilon = 1.5 \text{ w.p. } 0.5.$$

Compute $A(\mu_1)$ and $A(\mu_2)$ and compare them. (Briefly explain why the more dispersed distribution changes $A(\mu)$.)

Solution. (a) FOCs:

$$\alpha z \varepsilon k^{\alpha-1} n^\nu = r + \delta, \quad \nu z \varepsilon k^\alpha n^{\nu-1} = w.$$

Solve the ratio n/k from dividing the two:

$$\frac{\alpha}{\nu} \frac{n}{k} = \frac{r + \delta}{w} \Rightarrow \frac{n}{k} = \text{constant} \equiv \chi.$$

So $n = \chi k$. Plug into the first FOC:

$$\alpha z \varepsilon k^{\alpha-1} (\chi k)^\nu = \alpha z \varepsilon \chi^\nu k^{\alpha+\nu-1} = r + \delta.$$

Thus

$$k^{\alpha+\nu-1} = \frac{r + \delta}{\alpha z \chi^\nu} \cdot \varepsilon^{-1} \Rightarrow k^{1-\alpha-\nu} = \left(\frac{\alpha z \chi^\nu}{r + \delta} \right) \varepsilon.$$

Hence

$$k(\varepsilon) = \kappa_k \varepsilon^{\frac{1}{1-\alpha-\nu}}, \quad \kappa_k \equiv \left(\frac{\alpha z \chi^\nu}{r + \delta} \right)^{\frac{1}{1-\alpha-\nu}}.$$

And $n(\varepsilon) = \chi k(\varepsilon) = \kappa_n \varepsilon^{1/(1-\alpha-\nu)}$.

(b) Using the scaling from (a),

$$K = \int k(\varepsilon) d\mu = \kappa_k M(\mu), \quad N = \int n(\varepsilon) d\mu = \kappa_n M(\mu).$$

Aggregate output:

$$Y = \int y(\varepsilon) d\mu = \int z \varepsilon k(\varepsilon)^\alpha n(\varepsilon)^\nu d\mu.$$

Substitute $k = \kappa_k \varepsilon^a$, $n = \kappa_n \varepsilon^a$ where $a = 1/(1 - \alpha - \nu)$:

$$y(\varepsilon) = z \varepsilon (\kappa_k^\alpha \kappa_n^\nu) \varepsilon^{a(\alpha+\nu)} = z (\kappa_k^\alpha \kappa_n^\nu) \varepsilon^{1+a(\alpha+\nu)}.$$

But $1 + a(\alpha + \nu) = 1 + \frac{\alpha+\nu}{1-\alpha-\nu} = \frac{1}{1-\alpha-\nu} = a$. So $y(\varepsilon) = z (\kappa_k^\alpha \kappa_n^\nu) \varepsilon^a$. Integrate:

$$Y = z (\kappa_k^\alpha \kappa_n^\nu) M(\mu).$$

Now rewrite in terms of K, N . Since $K = \kappa_k M$ and $N = \kappa_n M$,

$$K^\alpha N^\nu = \kappa_k^\alpha \kappa_n^\nu M^{\alpha+\nu}.$$

Thus

$$zK^\alpha N^\nu A(\mu) = z(\kappa_k^\alpha \kappa_n^\nu) M^{\alpha+\nu} \cdot M^{1-\alpha-\nu} = z(\kappa_k^\alpha \kappa_n^\nu) M = Y,$$

with $A(\mu) = M^{1-\alpha-\nu}$.

(c) Here $1 - \alpha - \nu = 0.20 \Rightarrow a = 1/0.20 = 5$. So $M(\mu) = \mathbb{E}[\varepsilon^5]$ and $A(\mu) = M(\mu)^{0.20}$.

For μ_1 : $M(\mu_1) = 1^5 = 1 \Rightarrow A(\mu_1) = 1^{0.20} = 1$.

For μ_2 : $M(\mu_2) = 0.5 \cdot (0.5^5) + 0.5 \cdot (1.5^5)$. Compute: $0.5^5 = 1/32 = 0.03125$. $1.5^5 = 7.59375$. So $M(\mu_2) = 0.5(0.03125 + 7.59375) = 0.5(7.625) = 3.8125$. Then $A(\mu_2) = 3.8125^{0.20} \approx e^{0.20 \ln 3.8125} \approx e^{0.20 \cdot 1.338} = e^{0.268} \approx 1.307$.

So $A(\mu_2) > A(\mu_1)$. Reason: ε^a is convex for $a > 1$, so more dispersion (a mean-preserving spread) increases $\mathbb{E}[\varepsilon^a]$ by Jensen, raising $A(\mu)$ and thus aggregate output for given K, N .